Preliminary Investigation and Its Activities

1. Problem Identification and Definition

Many individuals, especially those new to investing, struggle to make informed decisions about where and how to invest their savings, such as 10 lakhs. The lack of personalized guidance, real-time market insights, and curated knowledge often leads to suboptimal investment choices, resulting in lower returns or higher risks.

2. Problem Description

Investors face challenges like information overload, difficulty understanding complex financial products, and lack of timely updates on market trends. Existing platforms may provide raw data but often fail to offer tailored advice or integrate multiple investment options in a user-friendly manner. This gap makes it hard for users to confidently allocate their funds across various instruments like stocks, mutual funds, fixed deposits, or bonds.

3. Fact Finding Techniques

3.1. Observation

Observing potential users interacting with current investment platforms reveals confusion due to complex interfaces and overwhelming data. Many users spend excessive time researching without clear direction, indicating a need for simplified, guided investment advice.

3.2. Interviews

Conducting interviews with novice investors, financial advisors, and platform users uncovers common pain points such as lack of trust in automated recommendations, difficulty

understanding jargon, and desire for personalized investment plans aligned with individual risk profiles and goals.

3.3. Questionnaires/Surveys

Surveys distributed among college students, young professionals, and middle-aged investors highlight that a significant percentage feel under-informed about investment options and express interest in a platform that offers curated knowledge, real-time insights, and easy-to-understand guidance.

3.4. Document Review

Reviewing financial reports, market analysis, and existing investment apps' documentation shows that while data availability is high, integration and personalization remain limited. Regulatory guidelines also emphasize the importance of investor education and transparency, which current systems partially address.

4. Drawbacks of Existing Systems

- Lack of personalized investment advice tailored to individual goals and risk tolerance.
- Overwhelming and complex user interfaces that intimidate beginners.
- Insufficient real-time insights and alerts to help users react promptly to market changes.
- Limited integration of diverse investment options in a single platform.
- Minimal educational content to help users understand investment fundamentals.

5. Scope for the Proposed System

The proposed Smart Investment Guide Platform aims to:

- Provide personalized investment recommendations based on user profiles and goals.

- Offer curated educational content to improve financial literacy.
- Integrate multiple investment options (stocks, mutual funds, fixed deposits, bonds) in one place.
- Deliver real-time market insights and alerts to enable timely decisions.
- Feature an intuitive, user-friendly interface suitable for beginners and experienced investors alike.

6. Feasibility Study

Technical Feasibility:

The platform can be developed using existing technologies such as data analytics, machine learning for personalized recommendations, and APIs for real-time market data. Cloud infrastructure can support scalability and security.

Economic Feasibility:

Initial development costs are moderate, with potential revenue streams from premium features, advisory services, or partnerships with financial institutions. The growing interest in personal finance apps suggests a viable market.

Operational Feasibility:

Users are increasingly comfortable with digital financial tools. The platform's educational focus and ease of use will encourage adoption. Continuous updates and customer support will ensure sustained engagement.

Legal Feasibility:

Compliance with financial regulations and data privacy laws will be ensured through proper legal consultation and adherence to guidelines.

Requirement Specification

1. Data Requirement of the System

The system requires various types of data to function effectively and provide meaningful investment guidance:

- **User Profile Data:** Personal details such as age, income, investment goals, risk tolerance, and investment horizon.
- **Market Data:** Real-time and historical data on stocks, mutual funds, bonds, fixed deposits, and other investment instruments. This includes prices, returns, volatility, and market news.
- **Financial Product Information:** Details about different investment options including interest rates, maturity periods, fund performance, fees, and terms & conditions.
- **Educational Content:** Curated articles, tutorials, FAQs, and glossaries to help users understand investment concepts.
- **Transaction Data:** Records of user investments, portfolio allocations, and transaction history for tracking and analysis.
- **Regulatory and Compliance Data:** Information related to financial regulations to ensure recommendations comply with legal standards.

2. Identify End Users of the System

The platform is designed to serve a diverse group of users, including:

- **Novice Investors:** Individuals with little or no prior investment experience seeking guidance to start investing their savings wisely.
- **Experienced Investors:** Users who want to optimize their portfolios with real-time insights and personalized recommendations.
- **Financial Advisors:** Professionals who may use the platform to assist clients with curated knowledge and data-driven advice.
- **Students and Learners:** Individuals interested in improving their financial literacy and understanding investment fundamentals.
- **Retirees and Senior Citizens:** Users looking for safe and stable investment options aligned with their retirement goals.

3. Input Data of the System

The system will accept the following inputs from users and external sources:

- **User Inputs:**
  - Personal information (age, income, financial goals)
  - Risk appetite (conservative, moderate, aggressive)
  - Investment amount (e.g., 10 lakhs)
  - Preferred investment duration and liquidity needs
  - Feedback and preferences on investment options
- **External Inputs:**
  - Real-time market feeds and price updates via APIs
  - Financial news and alerts
  - Updated product details from financial institutions
  - Regulatory updates and compliance rules

4. Output Information from the System

The platform will generate and present the following outputs to users:

- **Personalized Investment Plans:** Tailored recommendations based on user profile and market conditions.
- **Portfolio Suggestions:** Optimal allocation across various instruments like stocks, mutual funds, fixed deposits, and bonds.
- **Real-Time Alerts:** Notifications about market movements, investment opportunities, or risks.
- **Educational Insights:** Easy-to-understand articles, tips, and tutorials relevant to the user's investment journey.
- **Performance Reports:** Periodic summaries showing portfolio growth, returns, and risk metrics.
- **Comparative Analysis:** Side-by-side comparison of different investment options to aid decision-making.

5. Functional, Non-Functional, and Processing Requirements

5.1. Functional Requirements

- User registration and profile management.
- Risk assessment questionnaire to determine user risk tolerance.
- Integration with real-time market data providers.
- Algorithm to generate personalized investment recommendations.
- Dashboard to display portfolio status, alerts, and educational content.
- Search and filter options for exploring investment products.
- Secure transaction and investment tracking module.
- Feedback mechanism for users to rate recommendations and content.

5.2. Non-Functional Requirements

- **Usability:** Intuitive and user-friendly interface accessible on web and mobile devices.
- **Performance:** Fast response times for data retrieval and recommendation generation.
- **Scalability:** Ability to handle growing numbers of users and data volume without degradation.
- **Security:** Robust data protection, encryption, and compliance with privacy laws to safeguard user information.
- **Reliability:** High availability with minimal downtime and accurate data feeds.
- **Maintainability:** Modular design to allow easy updates and addition of new features.

5.3. Processing Requirements

- Real-time processing of market data to update recommendations and alerts promptly. • Batch processing for generating periodic performance reports and educational content updates.
- Data validation and error handling to ensure input accuracy and system stability.
- Machine learning or rule-based algorithms to analyze user data and market trends for personalized advice.

3. DATABASE DESIGN

3.1 Identify the Entities and the Attributes

USERS

-User ID

-Name

-Email ID

-Sign up

-Phone no

INVESTMENT

-Investment ID

-Initial Investment

-Investment Type

INFORMATION

-Information ID

-Strategy

-Description

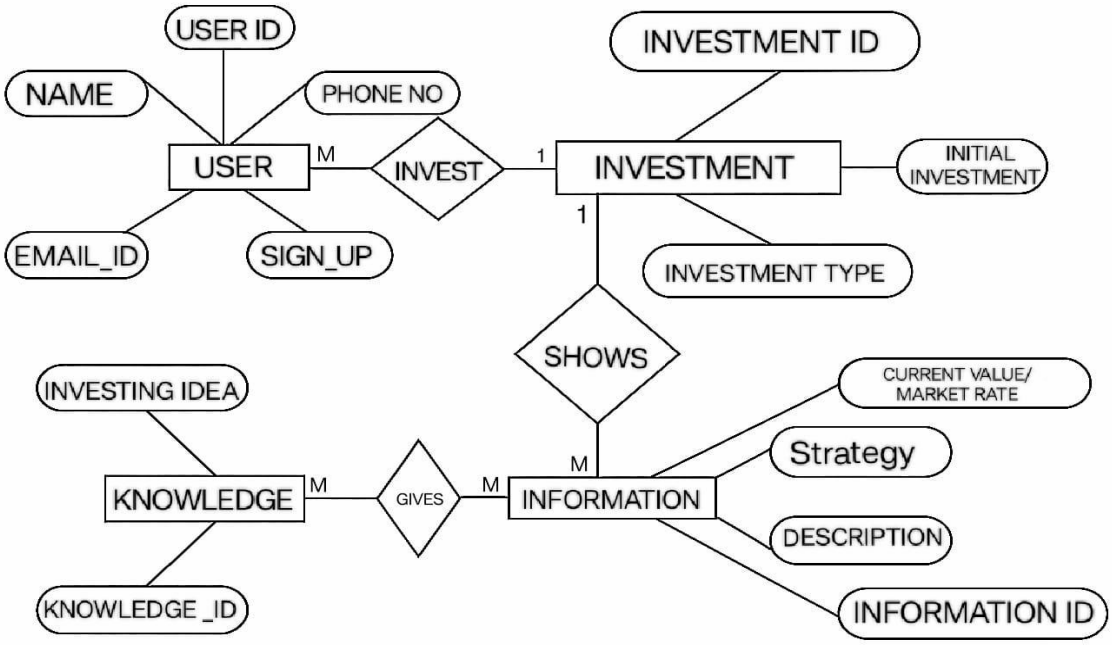
-Current value

KNOWLEDGE

-Knowledge ID

-Innvesting Idea

INVESTOPEDIA



3.3 Identifying all tables,fields and relationships

Table 1:Website

Field name	Data type	key	description
Name	Varchar(100)	Primary key	Website full name
URL	Varchar(50)	Unique key	Link of the website

Table 2:User

Field name	datatype	key	description
User ID	int	PK	Unique id for each user
Name	Varchar(50)		User's full name
Phone no	Varchar(50)		User's contact no
Email ID	Varchar(50)	unique	User's email id

Table 3:Investment

Field type	datatype	Key	description
Investment id	Int	PK	Unique id for each investment idea
Initial investment	float		Upto how much the user has to invest
Interest rate	percent		Interest rate of selected investment
Investment type	Varchar(100)		In which topic the user has to invest
Risk level	percent		Amount of risk in the investment.

Table 4: Information

Field name	Data type	key	Description
Information	int	PK	Unique id for each information provided
Current value	float		Current value of the following investment idea
Strategy	Varchar(100)		Strategy or ideas for investing
description	Varchar(100)		Details about the selected investment

Table 5: Knowledge

Field name	Data type	key	Description
Knowledge ID	Int		Unique ID for each knowledge
Investment ideas	Varchar(100)		How invest all knowledge is provided

3.4 Normalize Database

We assume attributes:-

-User ID

-User Name

-Investment Type

-Initial Investment

Take 5 sample record(UNNORMALIZED FORM-UNF)

User ID	Name	Investment Type	Initial Invesment
101	Akshay	Silver	170 , 20000
102	Ridhima	Real-Estate	50000
103	Kajal	Share Market , Silver	200 , 10000
104	Ritvik	Stocks	1000
105	Shanky	Mutual Funds	15000

- Solution:- 1NF (First Normal Form)

  Rule:- No repeating groups/multivalued attributes

  Each cell must hold atomic values.

User ID	User Name	Investment Type	Initial Investment
101	Akshay	Gold	20000
102	Ridhima	Silver	10000
103	Kajal	Real-estate	50000
104	Aishan	Stocks	1000
105	Shanky	Mutual funds	25000

- 2NF (Second Normal Form)

  Rule:- Must be in 1NF + No partial dependency (every non-key attribute must dependon whole primary key)

  Here ,composite key=(user ID , investment type selected) but attributes like user_name ,initial investment, depends only on user_Id not on investment type , so we split we split it into two tables.

  Table 1:- User

User_ID	User_name
101	Akshay
102	Ridhima
103	Kajal
104	Ritvik

2NF(Second Normal Form)

Rule: Must be in INF + No partial dependency(every non key attribute must depend on whole primary key)

Here,composite key=(User ID, investment type selected) but attributes like user _name,initial investment,depend only on user_id not on investment type,so we split it into two tables.

Table 1:User

User_id	User_name
101	Akshay
102	Ridhima
103	Kajal
104	Ritvik

Table 2 : Investment Type

Investment ID	User ID	Investment type	Initial Investment
01	101	Gold	20000
02	102	Silver	10000
03	103	Real-estate	50000
04	104	Stocks	1000
05	105	Mutual funds	25000

3NF(Third Normal Form)

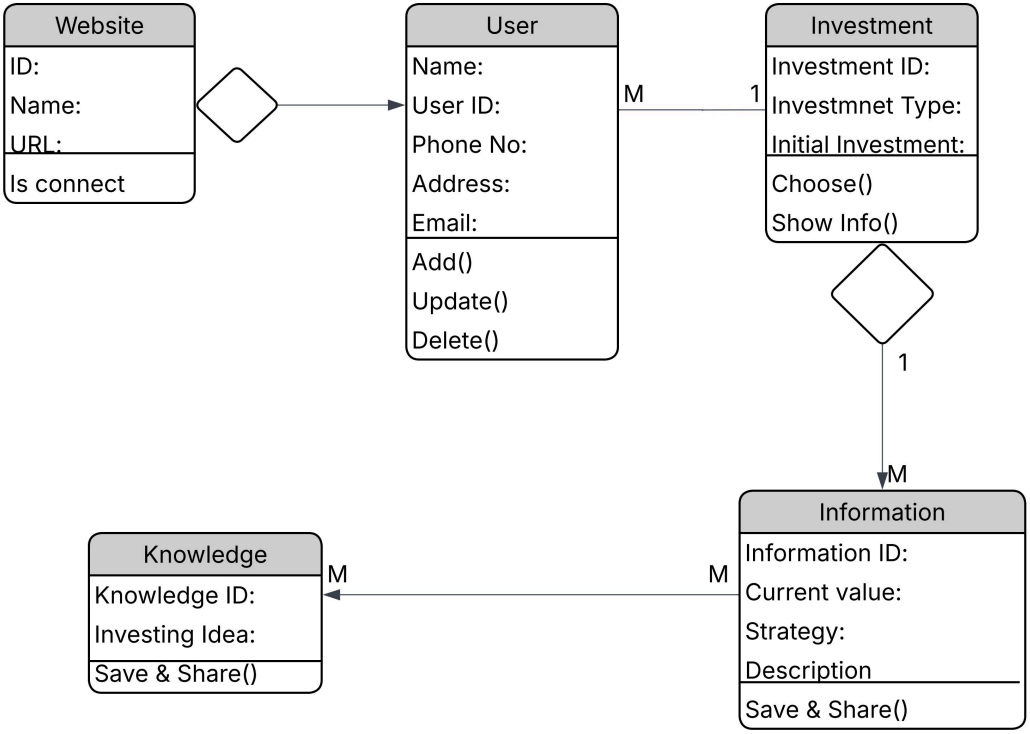
Rule:- Must be in 2NF+ No transitive dependency (non key attribute should not depend on another non key attribute)
In user table,User table,user_name depends only on udser_id

Investment ID directly depends on user ID
Non-transitive dependency here,but if we want to refine further,we can separate information details into another table.

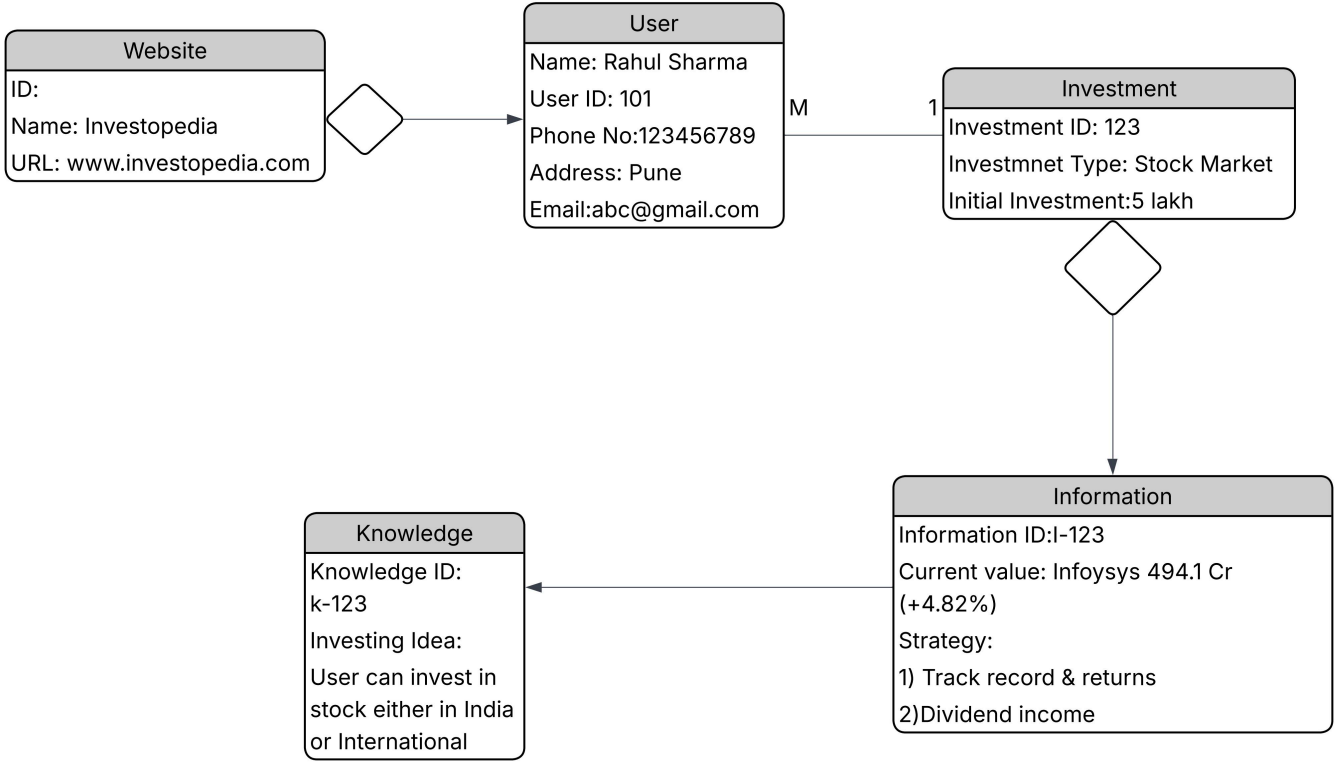
Table 3: Information

Investment ID	Info ID	Information
01	I1	Gold prices are high globally due to a safe haven demand.
02	I2	Real estate include a 30% drop in affordable housing supply.
03	I3	Stock market population/volume is decreasing day by day.

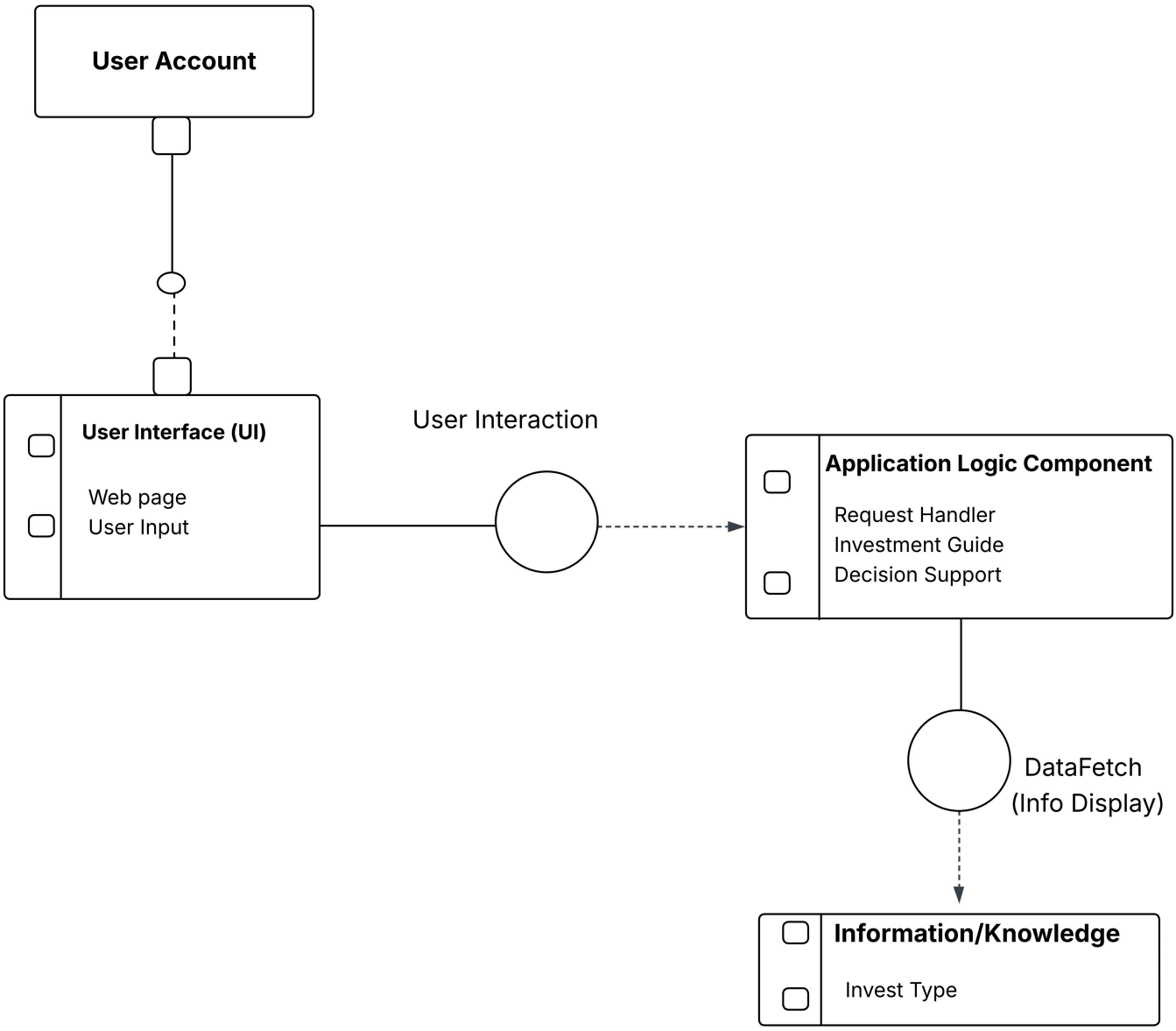
Class Diagram



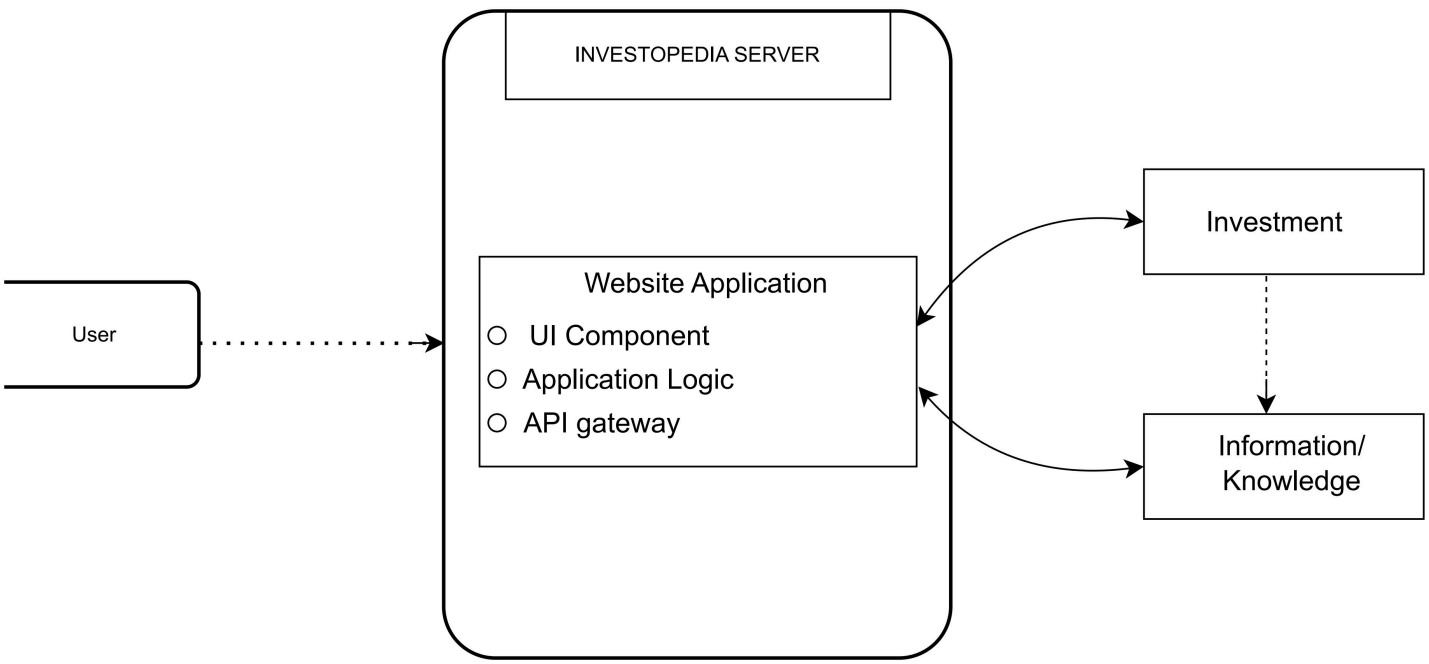
Object Diagram



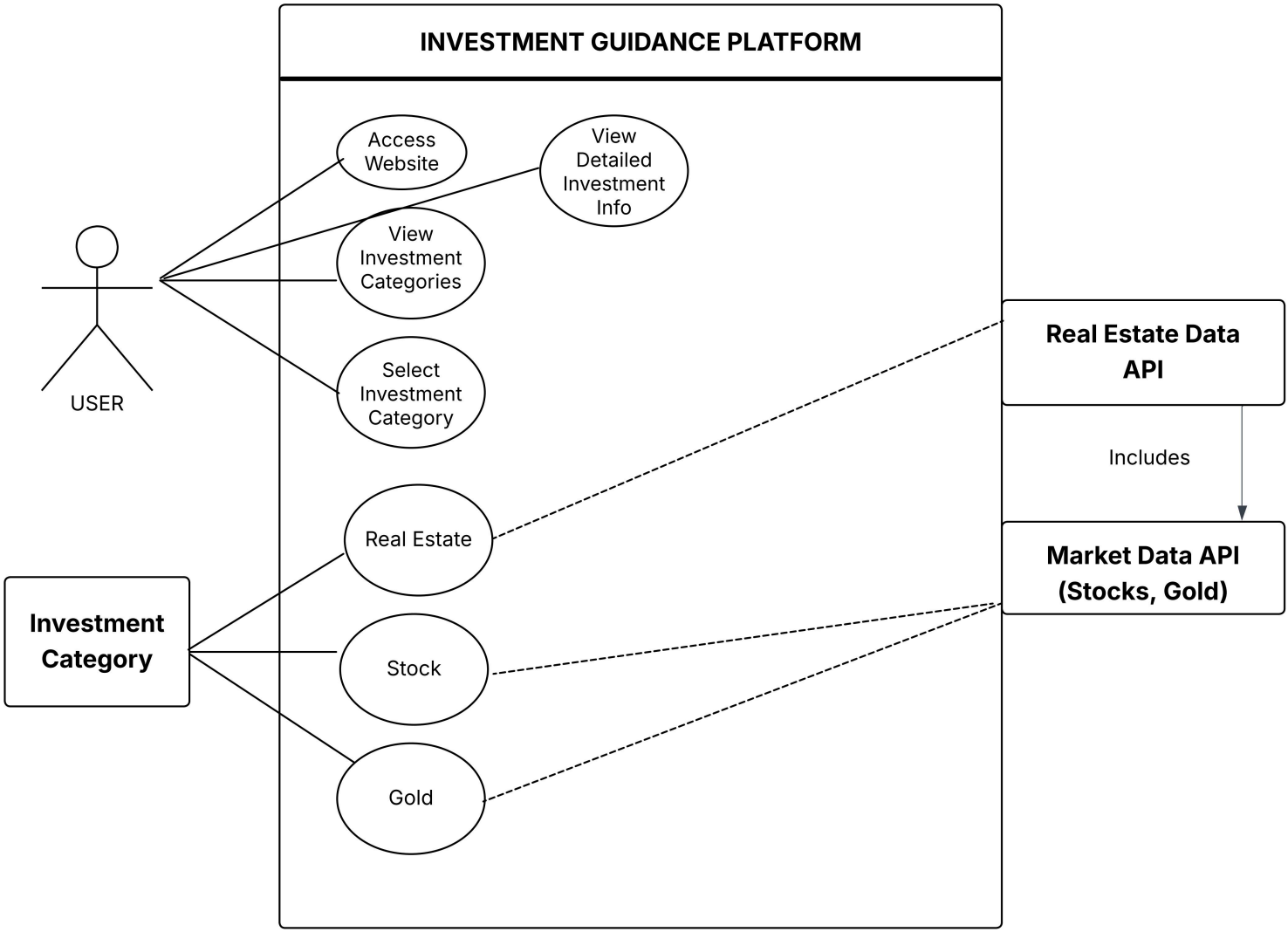
Component Diagram



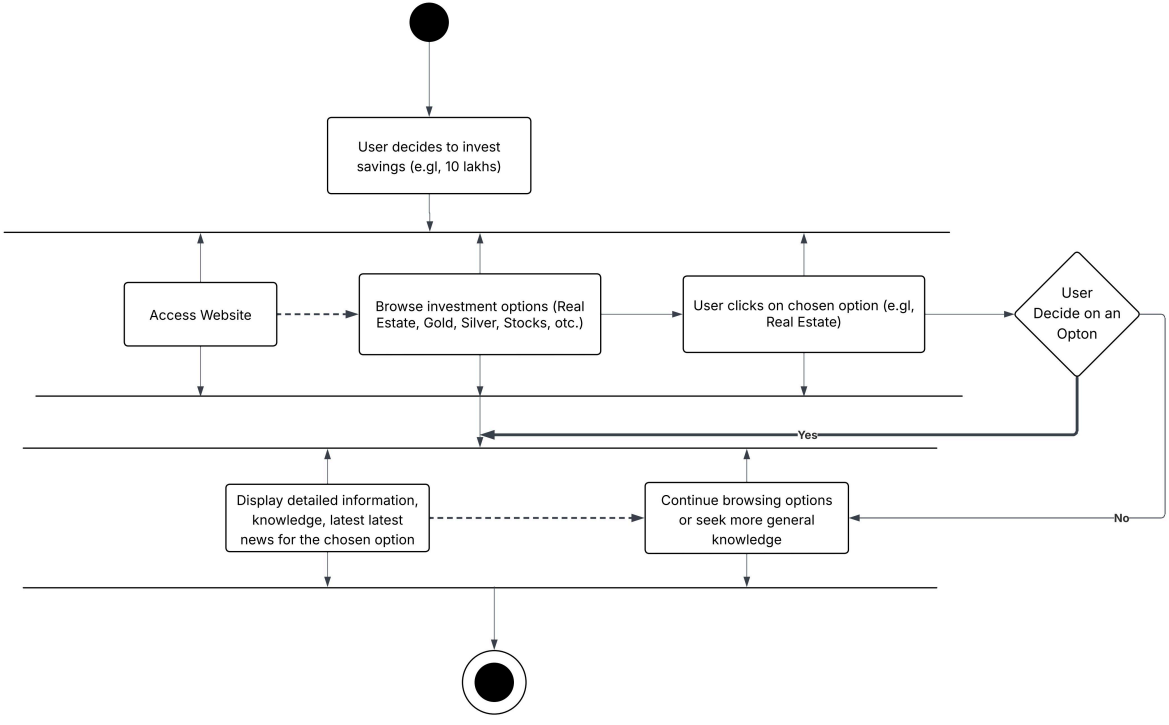
Deployment Diagram



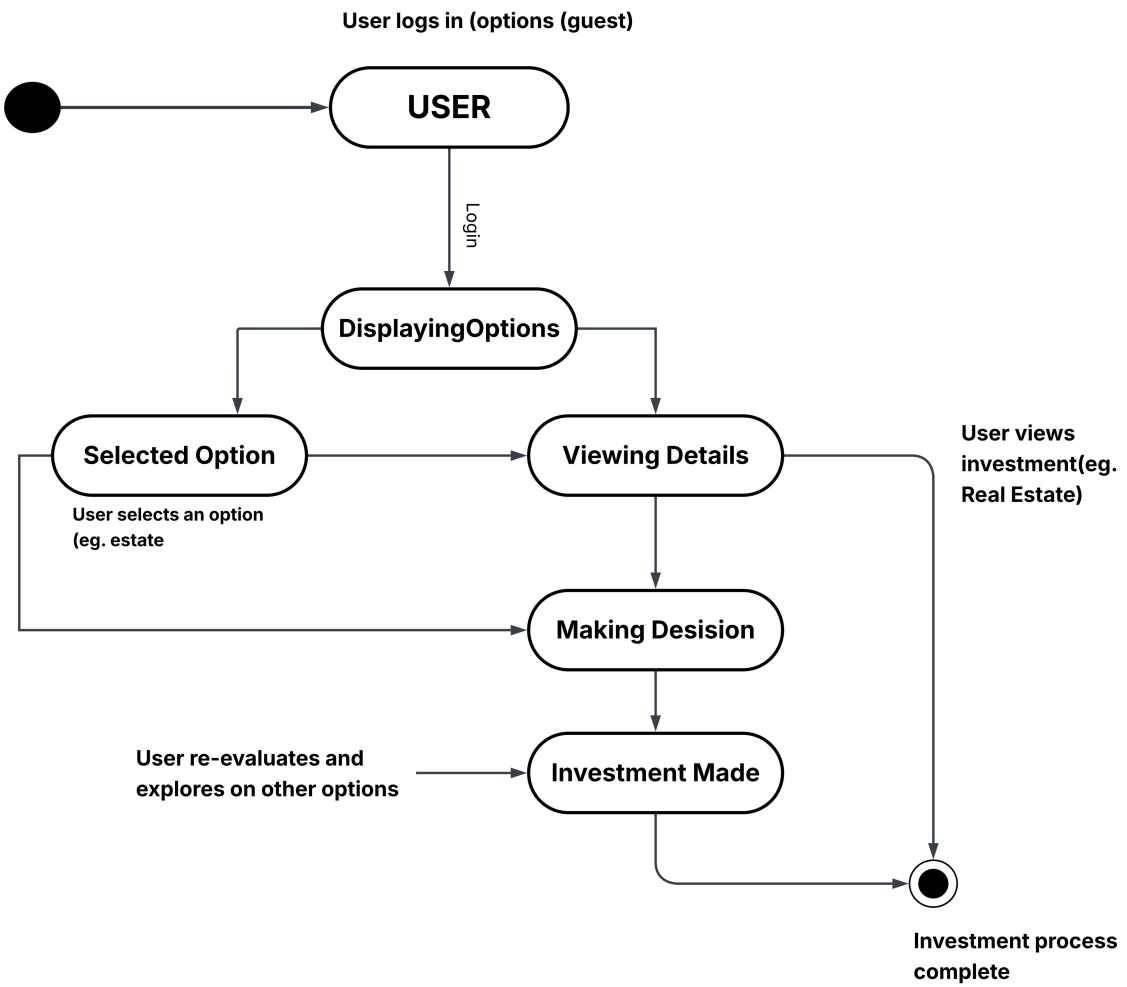
Use Case Diagram



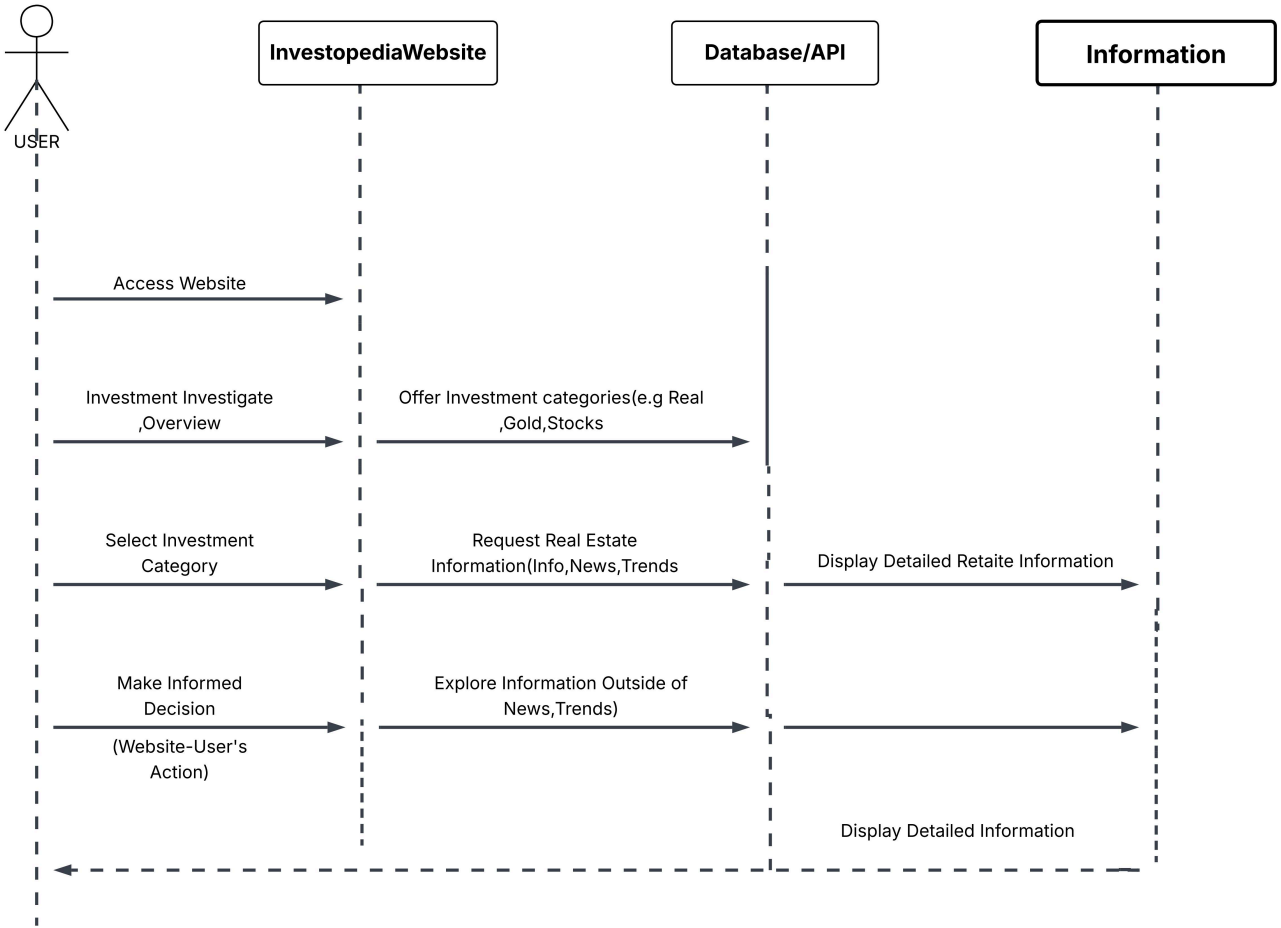
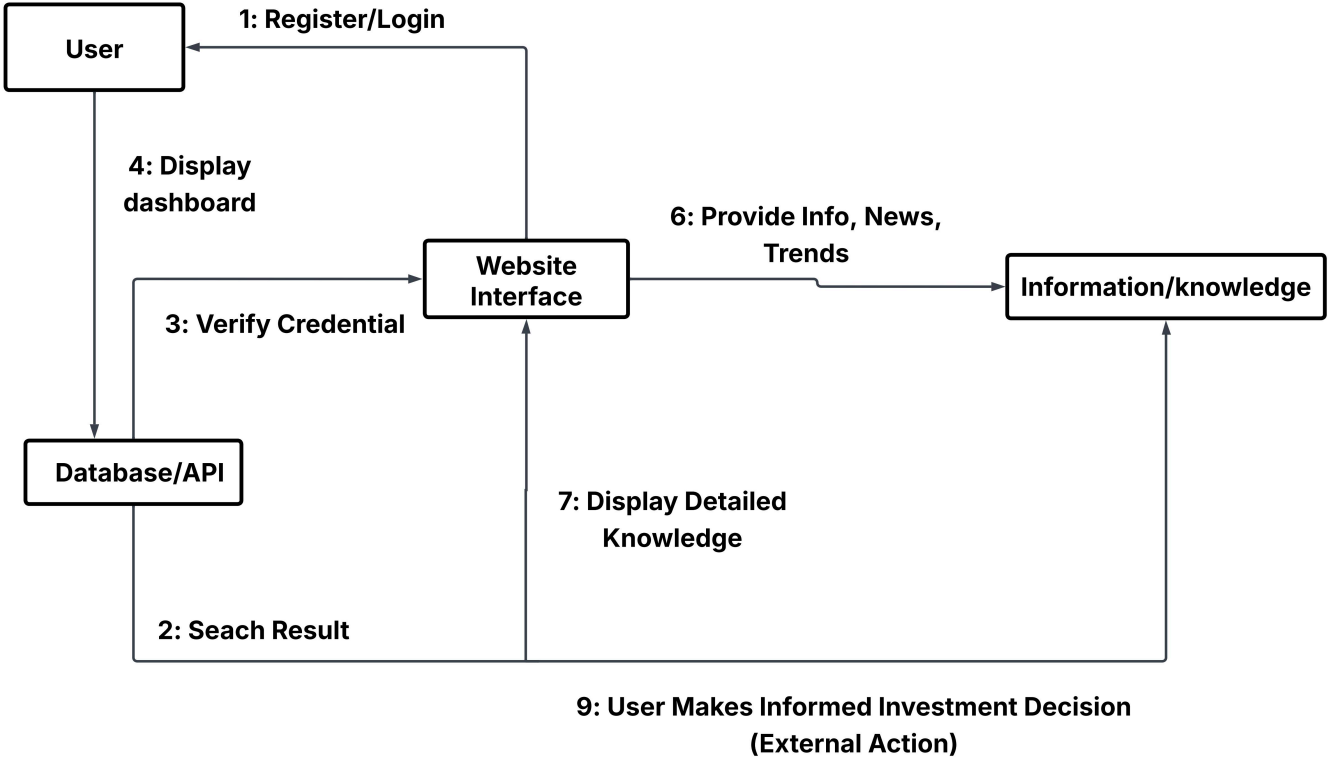
Activity Diagram



State Diagram



Sequence Diagram



Collaboration Diagram